

Problem Set 1, COSC330/PHYS306

Chapter 1 and Chapter 2-1 through 2-7

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1 Chapter 1

1. For the pulse shown in Figure 1–60, graphically determine the following:

- (a) rise time
- (b) fall time
- (c) pulse width
- (d) amplitude

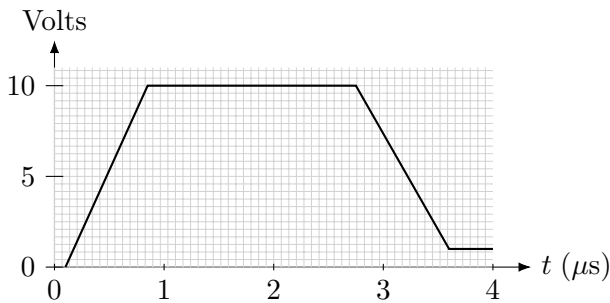


FIGURE 1–60

7. What is the total serial transfer time for the eight bits in Figure 1–62? What is the total parallel transfer time?
8. What is the period if the clock frequency is 3.5 GHz?

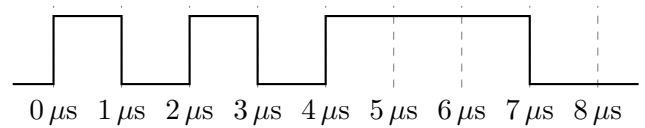


FIGURE 1–62

Section 1–3 Basic Logic Functions

9. Form a single logical statement from the following information:
 - (a) The light is ON if SW1 is closed.
 - (b) The light is ON if SW2 is closed.
 - (c) The light is OFF if both SW1 and SW2 are open.
10. A logic circuit requires HIGHS on all its inputs to make the output HIGH. What type of logic circuit is it?
11. A basic 2-input logic circuit has a HIGH on one input and a LOW on the other input, and the output is LOW. Identify the circuit.
12. A basic 2-input logic circuit has a HIGH on one input and a LOW on the other input, and the output is HIGH. What type of logic circuit is it?

2. Determine the period of the digital waveform in Figure 1–61.
3. What is the frequency of the waveform in Figure 1–61?
4. Is the pulse waveform in Figure 1–61 periodic or non-periodic?
5. Determine the duty cycle of the waveform in Figure 1–61.

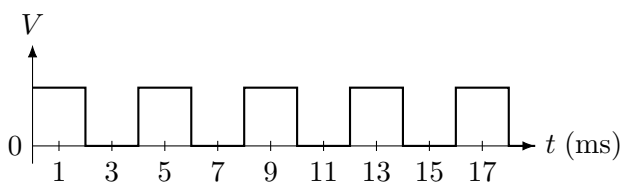


FIGURE 1–61

6. Determine the bit sequence represented by the waveform in Figure 1–62. A bit time is 1 μ s in this case.

2 Chapter 2

1. Convert each binary number to decimal:
 - (a) 110011.11
 - (b) 101010.01
 - (c) 1000001.111
 - (d) 1111000.101
 - (e) 1011100.10101

(f) 1110001.0001

2. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

- (a) two
- (b) three
- (c) four
- (d) five
- (e) six
- (f) seven
- (g) eight
- (h) nine
- (i) ten
- (j) eleven

3. Convert each number to binary using repeated division by 2:

- (a) 15
- (b) 21
- (c) 28
- (d) 34
- (e) 40
- (f) 59
- (g) 65
- (h) 73

4. Add the binary numbers:

- (a) $11 + 01$
- (b) $10 + 10$
- (c) $101 + 11$
- (d) $111 + 110$
- (e) $1001 + 101$
- (f) $1101 + 1011$

5. Express each decimal number as an 8-bit number in the 2's complement form:

- (a) +12
- (b) -68
- (c) +101
- (d) -125

6. Perform each addition in the 2's complement form:

- (a) $10001100 + 00111001$
- (b) $11011001 + 11100111$